

CSE370 : Database Systems

Practice Sheet

Concurrency Control

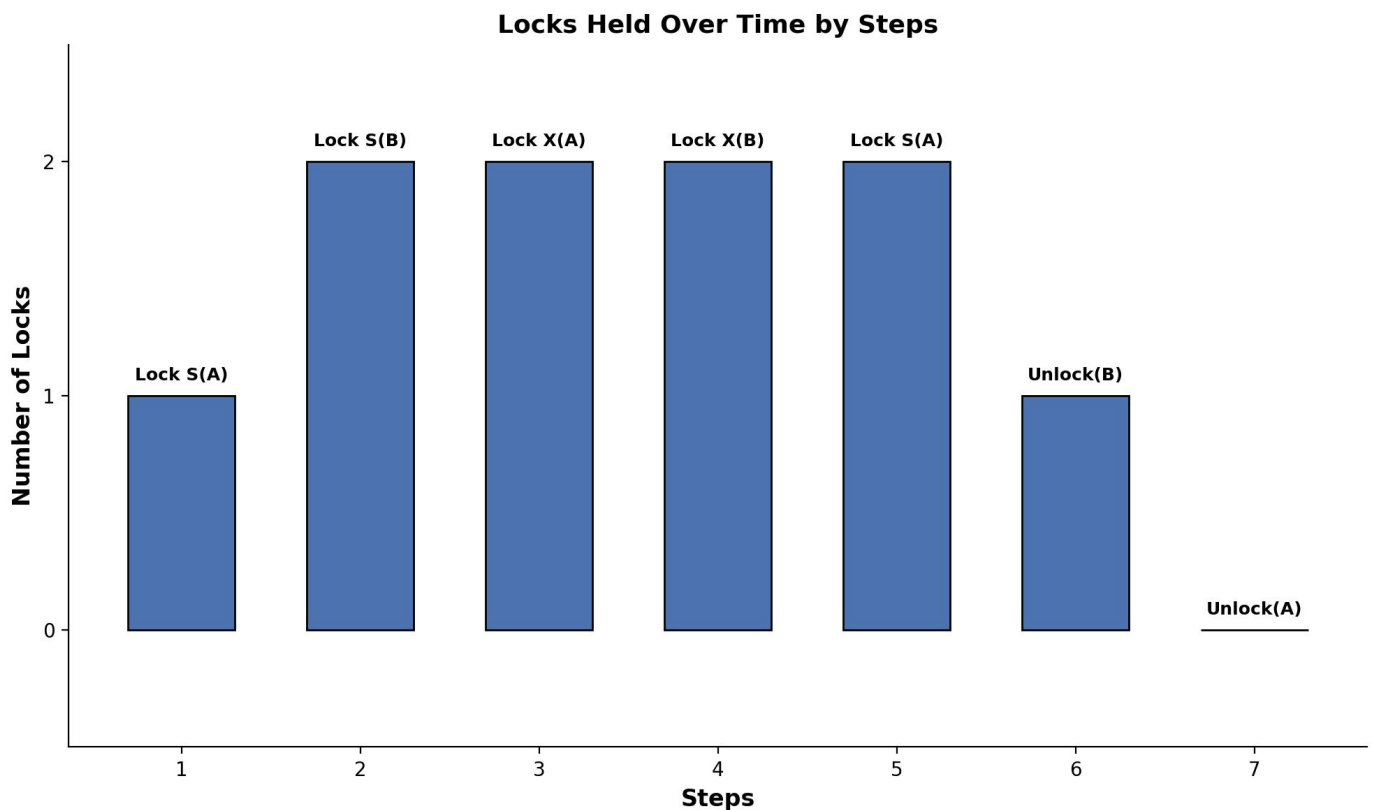
**Solutions are at the end of the document. SOLVE IT YOURSELF first.*

Question 1:

For the given graph below, two data items: A and B are being used.

Answer the following based on the graph:

- Identify and Explain the Lock Point here. (Mention Step number)
- Identify the starting and ending point of the Growing Phase. (Mention Step numbers)
- Identify the starting and ending point of the Shrinking Phase. (Mention Step numbers)
- Explain which locks have been upgraded. (Mention the data item and steps during upgrade)
- Explain which locks have been downgraded. (Mention the data item steps during downgrade)



Question 2:

State any two necessary conditions for a deadlock to occur.

Question 3:

What is the main purpose of deadlock prevention techniques in a database system? Name two timestamp-based deadlock prevention methods.

Question 4:

Explain the Wait-Die method. What happens when an older transaction requests a lock held by a younger transaction? What happens in the reverse case?

Question 5:

Explain the Wound-Wait method. What happens when an older transaction requests a lock held by a younger transaction? What happens in the reverse case?

Question 6:

State one major difference between Wait-Die and Wound-Wait.

Question 7:

Transaction T1 is older than T2.

T1 requests a lock currently held by T2. Instead of waiting, T1 causes T2 to be rolled back.

Which deadlock prevention method is being used?

Question 8:

Transaction T1 is younger than T2.

T1 requests a lock held by T2. The system immediately rolls back T1.

Which deadlock prevention method does this behavior represent?

Question 9:

T1 has timestamp 10 and T2 has timestamp 20.

T2 currently holds a lock on data item A. T1 requests the lock on A.

Under Wait-Die, should T1 wait or roll back?

Explain briefly.

Question 10:

T1 has timestamp 10 and T2 has timestamp 20.

T1 holds a lock on A. T2 requests the lock on A.

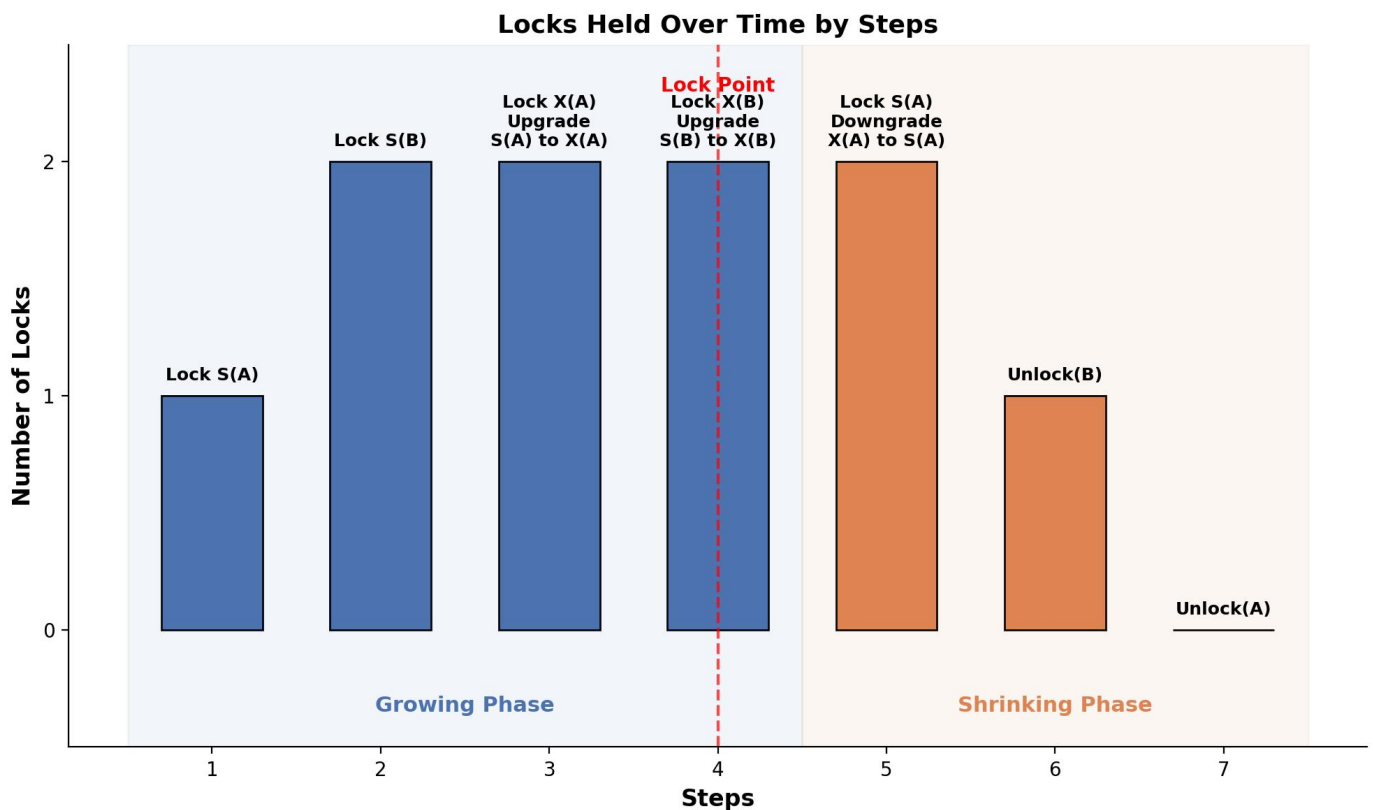
Under Wound-Wait, what happens?

SOLUTIONS

Answer 1:

- Identify and Explain the Lock Point here. (Mention Step number)
Lock Point: at step 4.
Explanation: Last step of growing phase, no more new locks or upgrades acquired after this step.
- Identify the starting and ending point of the Growing Phase. (Mention Step numbers)
Step 1 to 4.
- Identify the starting and ending point of the Shrinking Phase. (Mention Step numbers)
Step 5 to 7.
- Explain which locks have been upgraded. (Mention the data item and steps during upgrade)
Data item A upgraded on Step 3
Data item B upgraded on Step 4
- Explain which locks have been downgraded. (Mention the data item and steps during upgrade)
Data item A downgraded on Step 5

For better understanding a worked out graph is given below:



Answer 2:

hold and wait, no preemption, circular wait

Answer 3:

Prevent deadlock before it occurs; Wait-Die and Wound-Wait

Answer 4:

Older waits, younger dies/rolls back

Answer 5:

Older wounds/rolls back younger, younger waits

Answer 6:

Wait-Die allows older to wait; Wound-Wait allows older to preempt / force

Answer 7:

Wound-Wait

Answer 8:

Wait-Die

Answer 9:

Wait. T1 is older than T2

Answer 10:

T2 waits